# SOLUTION: Mock Exam – Digital Health & AI in Healthcare (WS 2025/2026)

Answers are given in blue

**Total: 52 points**
**Time: 60 minutes**

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## 1. AI Methods & Clinical Decision Support (14 points)

### 1.1 Matching Algorithms to Data Type and Reason (4 points)

Match each AI algorithm to the most appropriate:

- **Data type (List A)**
- **Reason (List B)**

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### List A – Data Types

A1. Discrete or categorical clinical outcomes
A2. High-dimensional imaging or omics data
A3. Small sample size datasets
A4. Longitudinal or time-series patient data
A5. Probabilistic relationships among variables
A6. Structured tabular EHR data

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### List B – Reasons

B1. Captures long-term dependencies with memory cells
B2. Encodes uncertainty via conditional probability distributions
B3. Learns compressed latent representations
B4. Performs well using prior assumptions to improve sample efficiency
B5. Models categorical likelihoods for classification
B6. Handles mixed numerical and categorical predictors efficiently

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### Algorithms

1. LSTM
2. Bayesian Network
3. Variational Autoencoder
4. Naïve Bayes

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### Answer:

LSTM: A4, B1
Bayesian Network: A5, B2
Variational Autoencoder: A2, B3
Naive Bayes: A1, B5

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# 1.2 LSTM Data Table Completion (4 points)

You are given hourly glucose measurements processed for an LSTM model.

Rule:
**Hyperglycemia = 1 if (Glucose ≥ 180 AND Mask = 1), else 0**

Δt = time since last valid measurement (Mask = 1).

| Time | Glucose | Mask | Δt | Hyperglycemia |
|---|---|---|---|---|
| 08:00 | 165 | 1 | 0 | [A] A = 0 |
| 09:00 | NaN | 0 | [B] | 0 B = 1 |
| 10:00 | 190 | 1 | 0 | [C] C = 1 |
| 11:00 | NaN | 0 | [D] | 0 D = 1 |
| 12:00 | NaN | 0 | [E] | 0 E = 2 |
| 13:00 | 210 | 1 | 0 | [F] F = 1 |

Fill in A–F.

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## 1.3 Evaluation Metric Selection (3 points)

For each scenario, select the **most appropriate metric** and briefly justify (1 sentence each).

Metrics to choose from:

- Accuracy
- Recall (Sensitivity)
- Precision (PPV)
- F1 Score
- AUC-ROC
- AUPRC

a) Screening tool for a rare but fatal disease

Recall because you don't want to miss false negatives

b) Confirmatory test before prescribing a toxic treatment

Precision because you don't want false positive before giving a risky treatment

c) Comparing two ML models across all classification thresholds

AUC-ROC because threshold-independent ranking measure

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## 1.4 Short Answer (3 points)

a) Name two challenges of AI in clinical decision support systems.

Data heterogeneity and missing data, lack of interpretability, dataset bias, regulatory and ethical concerns

b) What is the difference between overfitting and underfitting?

Overfitting: Model memorizes training data and performs poorly on other data
Underfitting: Model is too simple to capture underlying patterns

c) Why are external validation cohorts important in medical AI?

External validation ensures generalizability to new populations and prevents overly optimistic performance estimates

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# 2. Digital Biomarkers (8 points)

## 2.1 Time-Series in Healthcare (2 points)

List four examples of time-series data in healthcare.

Heart rate over time, blood measurements, continuous glucose monitoring, ICU vital signs

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## 2.2 Traditional vs Digital Endpoints (3 points)

Complete the table by assigning each item to:

- Endpoint
- Clinical Outcome
- Biomarker

| Item | Category |
|---|---|
| Wearable-measured step count variability | Biomarker |
| HbA1c blood test | Biomarker |
| Patient-reported depression questionnaire score | Clinical Outcome |
| Continuous heart rate variability from smartwatch | Biomarker |
| MRI-based tumor volume | Endpoint |

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## 2.3 Mental Health Digital Biomarkers (3 points)

Name three potential digital biomarkers for depression or anxiety derived from smartphone or wearable data and briefly state what behavioral signal they reflect.

GPS movement radius -> social withdrawal
Voice tone changes -> emotional state
Sleep duration variability -> insomnia

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# 3. Digital Twin (14 points)

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## 3.1 Conceptual Questions (4 points)

a) Define a Digital Twin in healthcare.

A digital twin is a computational model representing an individual patient that integrates data and simulations to predict disease progression or treatment response

b) What is the difference between a mechanistic model and a data-driven model?

Mechanistic: Based on biological/physical laws
Data-driven: Learns patterns from data without explicit domain equations

c) Why is population scaling important in Digital Twin models?

Population scaling allows models to generalize from individual simulations to broader populations

d) Name one limitation of Digital Twins in clinical practice.

High data requirements, computational cost, model uncertainty, validation difficulty

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## 3.2 Agent-Based Model Components (4 points)

Match each category to one definition (List A) and one example (List B).
Do not reuse items.

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### List A – Definitions

A1. Autonomous entities with behavioral rules
A2. Spatial or network structure where interactions occur
A3. System-wide parameters shared by all entities
A4. Emergent population-level outcome

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### List B – Examples

B1. Individual vaccination status
B2. Transmission rate parameter
B3. Social contact network graph
B4. Epidemic curve over time

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| Category | Definition | Example |
|---|---|---|
| Agent | A1 | B1 |
| Environment | A2 | B3 |
| Global Variable | A3 | B2 |
| Emergent Output | A4 | B4 |

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## 3.3 DE vs ABM Model Choice (3 points)

For each scenario, choose:

- Differential Equation (DE) Model
- Agent-Based Model (ABM)

Provide one concise reason.

a) Modeling national-level influenza prevalence trends.

DE because suitable for aggregate homogeneous population dynamics

b) Studying infection spread in a small school with heterogeneous contact patterns.

ABM because captures individual-level interactions and heterogeneity

## 3.4 SEIR Model Matching (3 points)

Match each compartment (S, E, I, R) to:

- A description (List A)
- A governing parameter (List B)

Do not reuse descriptions.

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### List A – Descriptions

A1. Infected and infectious individuals
A2. No prior exposure, susceptible to infection
A3. Infected but not yet infectious (latent)
A4. Recovered with immunity

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### List B – Parameters

B1. Transmission rate (β)
B2. Incubation rate (σ)
B3. Recovery rate (γ)

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**Answer:**

S: A2, B1
E: A3, B2
I: A1, B3
R: A4

# 4. Model-Guided Medicine (8 points)

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## 4.1 Bayesian Networks (4 points)

a) What is a Bayesian Network?

A probabilistic graphical model representing variables and their conditional dependencies using a directed acyclic graph

b) What does an edge represent?

Conditional dependency between variables

c) Why are Bayesian Networks useful in clinical decision-making?

They allow probabilistic reasoning and decision support under uncertainty

d) What is meant by inference under uncertainty?

Inference under uncertainty means calculating posterior probabilities given incomplete or uncertain evidence

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## 4.2 Hierarchical Models (2 points)

A statistical model that organizes parameters at multiple levels (e.g., patient, hospital, region) is called a hierarchical (multilevel) model.

Explain in one sentence why such models are useful in healthcare data.

They allow for clustered or nested data (e.g. patients within hospitals), improving estimation accuracy and capturing group-level effects

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## 4.3 Wisdom-Oriented Healthcare (2 points)

What distinguishes wisdom-oriented healthcare from purely data-driven prediction systems?

Wisdom-oriented healthcare integrates data, models, domain knowledge, and clinical context to support meaningful decision-making rather than only producing predictions

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# 5. Infrastructure, Standards & Regulations (8 points)

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## 5.1 IT Infrastructure (3 points)

Name three components required for implementing an AI-based decision support system in a hospital IT environment.

EHR integration, secure storage, user interface/dashboard

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## 5.2 DiGA and Regulation (3 points)

a) What is a DiGA?

DiGA (Digitale Gesundheitsanwendung) is a certified digital health application that can be prescribed and reimbursed in Germany.

b) Name two requirements for regulatory approval of digital health applications in Europe.

Clinical evidence of benefit, data protection

c) Why is evidence generation important for reimbursement?

Evidence is required to demonstrate safety, efficacy, and added healthcare value for reimbursement decisions

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## 5.3 Data Governance (2 points)

a) What is the purpose of anonymization or pseudonymization in medical datasets?

Protects patient privacy and ensures GDPR compliance

b) Why is interoperability (e.g., HL7 FHIR) important for digital health systems?

Interoperability allows standardized healthcare exchange across systems